EE3032 Introduction to VLSI Design—Final

Fall 2009

Reconstructed from the complete three-page original (printed pages 1-1 through 1-3). The seven printed question weights sum to **90%**; no missing 10% question is invented. Source wording, the full overbar in Q7, and all transistor/clock polarities are preserved. All figures are editable TikZ, including paths from the native vector layout.

1. (15%) Figure 1 shows a circuit for realizing a low-cost latch. Complete the following questions.

(a) What type of latch is realized by the circuit? (5%)

(b) Assume that the switching threshold voltage of the inverter Inv1 is $V_{dd}/2$. List two inequalities for determining the sizes of P2 and N2. (10%)

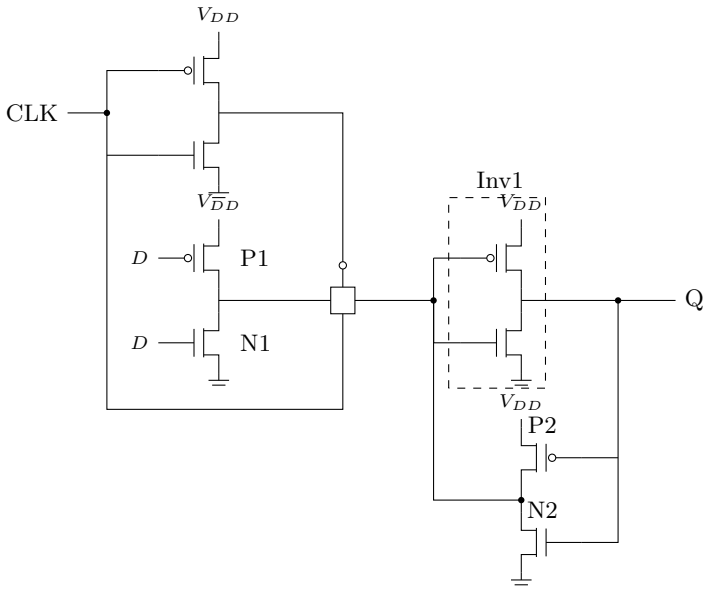


Figure 1: A low-cost latch. The feedback pair is always connected.

2. (15%) Explain the operation principle of the circuit shown in Figure 2 shows a circuit for realizing a dynamic register. Answer the following questions.

(a) Explain the operation principle of the circuit. (5%)

(b) Design the register using the static low-cost latches shown in Figure 1. (10%)

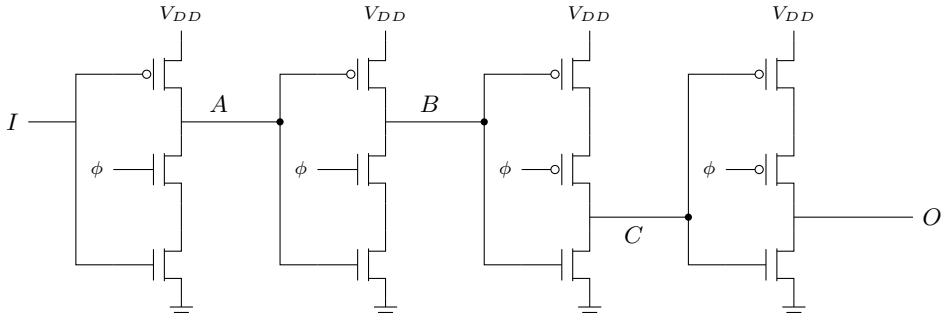


Figure 2: A dynamic CMOS register circuit. All four three-device stages and the two clock-controlled pMOS bubbles are retained.

3. (10%) Complete the following questions.

(a) Figure 3(a) shows a three-input dynamic logic gate. Assume that the sizes of the NMOS and PMOS transistors of the dynamic logic gate and the inverter are equal and $C_g/C_d = \alpha$ ($C_d = C_s$). Also, the threshold voltage of the inverter is $V_{dd}/2$. Derive the minimal value of α such that the circuit can work correctly. (5%)

(b) Complete the restore circuit of Figure 3(b), such that $Y = 0$ when $X = 0$ and $C = 0$. (5%)

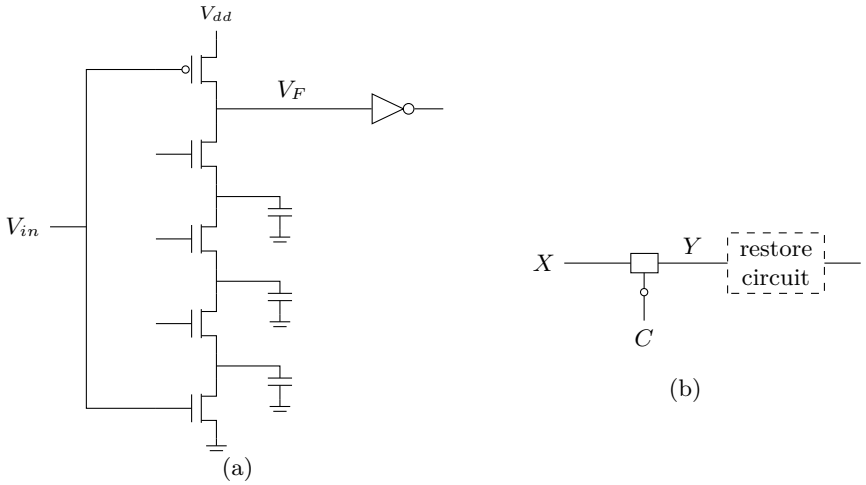


Figure 3: (a) Dynamic logic. (b) Pass-transistor design. The unlabelled three data gates in (a) are preserved.

4. (15%) Figure 4 shows a circuit designed with NP-domino logic. Complete the following questions.

(a) What function does this circuit realize? (5%)

(b) Using timing diagrams, describe the operation of the circuit. (5%)

(c) Design the circuit using Domino logic. (5%)

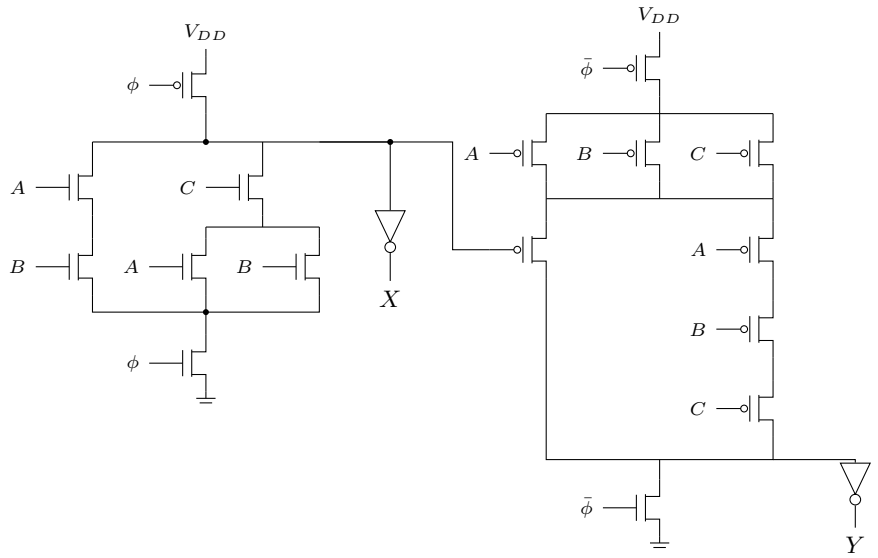


Figure 4: A circuit designed with NP-domino logic. The static output inverters X and Y are retained.

5. (10%) Figure 5 shows a sequential circuit (pipelined circuit). Assume that your boss asks you to design the circuit such that the circuit can be operated at $200\,MHz$. Also, assume that the clock-to-Q propagation delay, the clock-to-Q contamination delay, the setup time, and the hold time of registers R1 and R2 are $0.5\,ns$, $0.2\,ns$, $1.5\,ns$, and $1\,ns$, respectively. Determine the maximum and minimum delays of the combinational circuit.

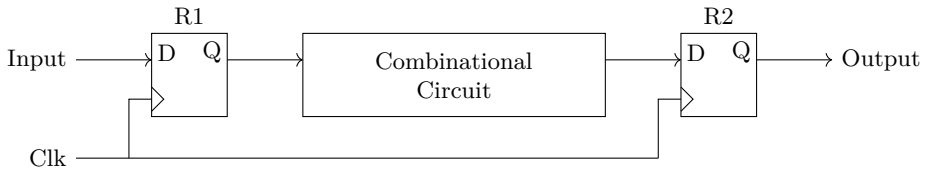


Figure 5: A pipelined circuit.

6. (10%) Figure 6 shows the symbolic layout of a circuit. Draw the transistor-level schematic diagram of the circuit.

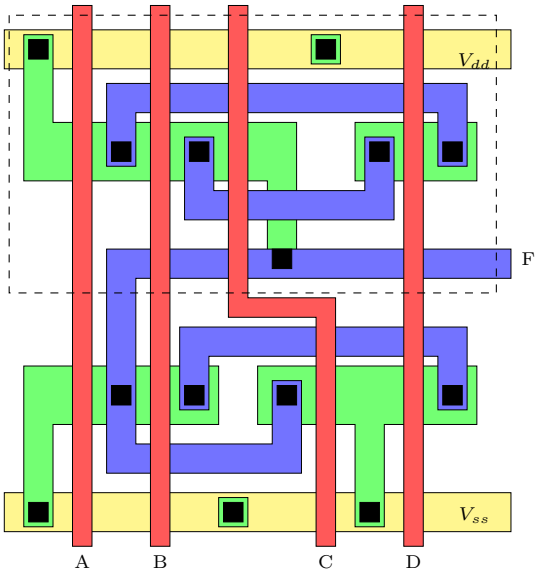


Figure 6: The symbolic layout. Editable native path coordinates: green diffusion, yellow Metal 2, blue Metal 1, red poly, black contact; dashed n-well. Crossings without black contacts do not connect layers.

7. (15%) Sketch the Boolean function $f = \overline{AB + C + DE}$ using each of the following circuit techniques. (1) Domino logic (5%) (2) Fully differential logic (5%) (3) Static CMOS logic (5%).